

# Tim Perr

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## Profile

Graduate student passionate about scientific computing and exploring the intersection of computing, astronomy, biology, and physics. I enjoy tackling complex problems, working with data, and diving into systems-level programming, operating systems, and AI related projects. I'm excited about contributing to projects where science, technology, and creativity come together to make an impact.

## Education

### Master of Science in Computer Science

Michigan Technological University Aug 2024–Dec 2025

### Master of Science in Data Science

Michigan Technological University Aug 2024–Dec 2025

### Bachelor of Science in Computer Science

Michigan Technological University Aug 2021–Apr 2024

## Skills

**Programming Languages:** Python, C/C++, Java, Swift, R, Matlab, Bash, MIPS Assembly, Makefile, LaTeX

**Web Development:** React.js, Node.js, HTML, CSS, JavaScript, Flask, PHP, APIs

**Data and Machine Learning:** SQL, Data Analysis, Machine Learning, Python libraries (e.g., NumPy, Pandas, Matplotlib), MATLAB Toolkits

**Systems and Theoretical Knowledge:** Concurrent/Parallel Computing, Quantum Computing, Discrete Mathematics, Data Structures, Programming Language Design, Computing Theory

**Mathematics and Science:** Linear Algebra, Calculus, Statistics, General Physics and Astronomy

**Artificial Intelligence:** Various AI methods, AI in Healthcare, AI in Climate Science, Computer Vision

**Other Related Skills:** Linux, Git, Agile, basic Cyber Skills, Docker

## Work and Research Experience

### Computer Science Research Assistant

Michigan Technological University Feb 2023 – July 2025

- Contributed to the development of a website for Illuminated Devices, supported by an NSF-funded project [#BCS-2122034](#)
- Presented research on the theoretical aspects of the project at ASEE 2024; view the paper <https://peer.asee.org/46774>

### Spinel Lab Member

Michigan Technological University Nov 2024 – Present

- Focused on Computer Science Education, specifically in reframing perceptions of computing and developing supplemental learning materials
- Responsible for data analysis and rewriting homework questions to enhance learning materials

### Computer Science Learning Center Coach

Michigan Technological University Sep 2022 – Apr 2024

- Guided students through coursework across various levels, from introductory to graduate computing courses

### Computer Science Lab Assistant

Michigan Technological University Sep 2022 – Dec 2022

- Assisted in teaching introductory computer science labs (CS1111, CS1131, CS1122)

### Instructor

iD Tech Camps

May 2022 – Aug 2022

- Delivered curriculum to classes of up to 12 students, ensuring engagement and understanding of programming concepts
- Monitored students' progress toward learning goals and facilitated non-instructional activities

### Bank Teller

First National Bank of LaGrange

Oct 2018 – Aug 2021

- Processed cash transactions, balanced drawers and vaults, issued cashier's checks, and delivered exceptional customer service

## Projects

### Entropy Emulator

*Objective:* Develop a simple emulator for a computer

- Programmed in C++ utilizing a non-complex Makefile
- Reads a file of computer operations (clock tick, memory load, CPU load) and executes commands with assembly-like operations
- Includes operations such as Load/Save Word and Branches
- <https://github.com/tperr/Entropy-Emulator>

### Fracture Detector

*Objective:* Use a neural network to detect fractures

- Final project for SAT5114: AI in Healthcare
- Implemented in Python using U-Net architecture
- Trained on a dataset provided by [figshare.com](https://figshare.com)
- <https://github.com/tperr/Fracture-Detector>

## Certificates

### Essentials of Free and Open Source Software

Canvas Credentials (Badgr)

08/2024

<https://api.badgr.io/public/assertions/EXNMMFfHTc27lXLTj2oS1w>

### Deep Learning Onramp

Mathworks

11/2024

<https://matlabacademy.mathworks.com/progress/share/certificate.html?id=ee0531f0-bb8a-408c-8604-22fd76a5783d&>